

1755. Closest Subsequence Sum

Hard

Topics

Companies

Hint

You are given an integer array `nums` and an integer `goal`.

You want to choose a subsequence of `nums` such that the sum of its elements is the closest possible to `goal`. That is, if the sum of the subsequence's elements is `sum`, then you want to **minimize the absolute difference** `abs(sum - goal)`.

Return the **minimum** possible value of `abs(sum - goal)`.

Note that a subsequence of an array is an array formed by removing some elements (**possibly all or none**) of the original array.

Example 1:

Input: `nums = [5,-7,3,5]`, `goal = 6`

Output: 0

Explanation: Choose the whole array as a subsequence, with a sum of 6. This is equal to the goal, so the absolute difference is 0.

Example 2:

Input: `nums = [7,-9,15,-2]`, `goal = -5`

Output: 1

Explanation: Choose the subsequence `[7,-9,-2]`, with a sum of -4. The absolute difference is `abs(-4 - (-5)) = abs(1) = 1`, which is the minimum.

Example 3:

Input: `nums = [1,2,3]`, `goal = -7`

Output: 7

Constraints:

- `1 <= nums.length <= 40`
- `-107 <= nums[i] <= 107`
- `-109 <= goal <= 109`

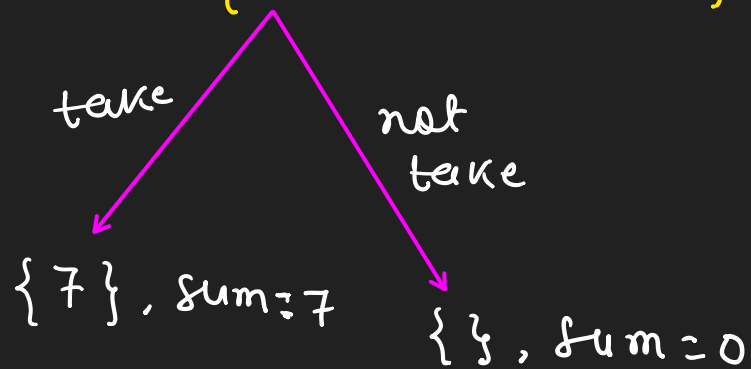
→ looking unusual

Thought Process

$\text{nums} = \{ \overset{0}{7}, \overset{1}{-9}, \overset{2}{15}, \overset{3}{-2} \}, \text{goal} = -5$

- Brute force

$\text{nums} = \{ \overset{0}{7}, \overset{1}{-9}, \overset{2}{15}, \overset{3}{-2} \}, \text{goal} = -5$



- options

↳ Recursion

$n \rightarrow 2^n$

$O(2^n) \rightarrow \text{TLE}$

↳ $(2^{40}) \rightarrow 10^{12} (\text{TLE})$

$\text{solve}(i, \text{sum})$

$\left. \begin{array}{cc} [n] & [\text{sum}] \\ 40 & * & 10^9 \end{array} \right\} \text{TLE}$

- Recursion

- D.p
 - ↳ memoization
 - ↳ Tabulation

} shows TLE

Meet in the middle

we can think

